

The Linguistic Cost of Low-Rank Compression: An SVD Analysis of Transformer Models for Part-of-Speech Tagging

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Abstract

Transformer models are increasingly becoming the dominant model for sequence-labeling tasks; however, their parameter counts grow rapidly with model depth and embedding dimension, restricting deployment on various devices with memory and other resource constraints. This is often tackled using Singular Value Decomposition (SVD), a widely studied compression technique that approximates large weight matrices by truncated low-rank factorizations. However, the linguistic consequences of this decomposition on syntactic tasks are not widely understood. In this paper, we apply truncated SVD to the attention weight matrices of three transformer-based Part-of-Speech (POS) taggers (448K, 1.68M, and 2.25M parameters) trained on the Wall Street Journal section of the Penn Treebank. We compress across various ranks $k \in \{1, 2, 4, 8, 16, 24, 32\}$, measuring the loss of train and test, the accuracy of the test, and the error patterns at the tag-level. Very aggressive compression (94–97% of attention parameters disabled) preserves accuracy within 0.4 points of the uncompressed baseline on the smaller, well-trained model, but mid-range ranks induce a non-monotonic accuracy collapse on under-trained larger models, dropping to 11.17% at $k = 16$ in the 2.25M model. Targeted error analysis confirms that compression disproportionately destroys fine-grained syntactic distinctions: Proper vs. Common Nouns (NNP $\leftrightarrow$ NN, 625 errors), past tense vs. past participle (VBD $\leftrightarrow$ VBN, 283 errors), and adjectives vs. adverbs (JJ $\leftrightarrow$ RB, 84 errors).

1 Introduction

Pre-trained transformer models such as BERT (Devlin et al., 2019) and its many descendants have rapidly become the standard architecture for nearly every Natural Language Processing (NLP) task, including sequence labeling tasks like Part-of-Speech (POS) tagging. However, this accuracy comes through a trade-off with memory — even relatively

small transformer encoders require megabytes of weights, while higher-precision models routinely need multiple gigabytes. This inhibits deployment on devices with fewer resources, such as mobile phones and embedded systems, where computation and storage experience high constraints.

Among the dominant strategies for transformer compression—pruning, quantization, knowledge distillation, and matrix factorization—Singular Value Decomposition (SVD) holds a special theoretical appeal. During SVD-based factorization, any weight matrix W can be approximated by the product of two narrower matrices whose combined parameter count is smaller than the original. Thus, by trading a controlled amount of reconstruction error, SVD produces a controlled reduction in parameters.

Yet despite its nuanced approach, SVD-based compression continues to be preferred for its ability to dissect computation efficiency rather than linguistic fidelity. Our paper differs here, answering not the question of how much smaller a transformer can be, but by focusing on what kinds of linguistic losses occur in a transformer when its weight matrices are compressed. To reiterate, we set out to determine (i) how truncated SVD applied to attention weight matrices affects POS tagging accuracy on the Wall Street Journal corpus, and (ii) which specific tag confusions become more frequent as the rank k of the factorization decreases.

Our tests showed that the relationship between rank and accuracy depends on the underlying model size and amount of training data. In our tests, we found that on a 448K-parameter tagger trained on 15K sentences, accuracy is essentially preserved across all tested ranks (85.86–86.31% vs. 86.34% baseline), even at 94.5% attention parameter disabling. On a 1.68M-parameter model trained on 1K sentences, very aggressive compression ($k = 1$, 96.35% disabled) retains 63.81% accuracy while moderate compression ($k = 8$) collapses accuracy

to 52.67%. On a 2.25M-parameter model trained on the same 1K sentences, an even more striking collapse appears at $k = 16$, where accuracy drops to 11.17%. Error analysis on the aggregate confusion set reveals that the most common compressed-model errors are precisely the syntactic distinctions that depend on rare or context-sensitive features: $\text{NNP} \leftrightarrow \text{NN}$, $\text{VBD} \leftrightarrow \text{VBN}$, and $\text{JJ} \leftrightarrow \text{RB}$.

2 Related Work

Our work builds on four lines of prior research: transformer compression, low-rank weight approximation, transformer interpretability, and POS tagging.

Transformer compression. Wang et al. (2020) shows that structured pruning and low-rank factorization can significantly reduce transformer size without large performance losses. Their work focuses on language modeling, while we study effects on a syntactic sequence labeling task. Abdaoui et al. (2020) compress multilingual BERT using pruning and distillation, providing a baseline for comparison. Unlike distillation, SVD is applied after training, highlighting a tradeoff between training cost and accuracy.

Low-rank decomposition. Raunak et al. (2019) apply SVD for extreme compression in NLU classification tasks, showing that aggressive truncation can still perform well. We test whether this holds for token-level tasks like POS tagging, where predictions depend on a more specific context.

Transformer internals. Rogers et al. (2020) surveys what attention heads learn, including syntactic features relevant to POS tagging. This suggests that compression can disproportionately affect rare words and modifier categories, which rely on lower-importance components.

Efficient attention. Lei (2021) reduces attention computation through recurrence, motivating our focus on compressing query, key, and value matrices. We also build on the original transformer architecture introduced by Vaswani et al. (2017).

POS tagging. The Wall Street Journal portion of the Penn Treebank (Marcus et al., 1993) is a standard benchmark for POS tagging. We use this to test and train our model

3 Methodology

3.1 Singular Value Decomposition

Transformer models represent relationships between words using large weight matrices $W \in \mathbb{R}^{d \times d}$, especially in the query, key, and value projections of the attention mechanism. Singular Value Decomposition (SVD) provides a way for the approximation of these large matrices using smaller components. Thus, any matrix W can be written as:

$$W = U \Sigma V^\top,$$

where U and V contain orthogonal vectors representing learned patterns, and Σ is a diagonal matrix whose entries (singular values) are ordered by importance.

To compress the model, we only retain the top k singular values and their corresponding vectors:

$$W \approx U_k \Sigma_k V_k^\top. \quad (1)$$

This reduces the number of parameters from d^2 to approximately $2dk$, allowing for the level of compression through k to be controlled. Note, the smaller the values of k , the more aggressive the compression.

In simpler terms, the largest singular values capture the dominant patterns in the data, while the smaller singular values are used to represent the subtle or rarer features. Given this, we hypothesize that reducing k will disproportionately remove fine-grained linguistic information, such as distinctions involving rare words, modifier categories, and context-dependent structures.

3.2 Hypotheses

We propose and test three predictions about how compression affects linguistic behavior.

Prediction 1: Rare and Proper Nouns (NNP). We expect compression to reduce precision on rare vocabulary, especially proper nouns. The largest singular values likely capture common syntactic patterns (e.g., frequent function words), while discarded values likely contain the nuanced parameters needed to identify rare vocabulary. So, we propose that as these smaller parameters are removed, the model may misclassify proper nouns (NNP) as common nouns (NN) or default to more frequent tags for the unknowns.

Prediction 2: Adjective vs. Adverb Confusion (JJ vs. RB). Adjectives and adverbs often require subtle contextual cues to distinguish as adjectives

(JJ) modify nouns, while the adverbs (RB) modify verbs and adjectives. We propose that as the rank k decreases, the model will blur these categories into a more general “modifier” representation, leading to increased confusion between JJ and RB tags.

Prediction 3: Context-Dependent Grammar. Transformers rely on attention weight matrices to capture relationships across a sentence. We hypothesize that heavy compression will weaken this ability, making it harder to resolve context-dependent cases. In particular, the model may fail on tokens whose POS tags rely heavily on surrounding syntactic structures to be resolved, such as auxiliary verbs needed for verb form distinctions (e.g., VBD vs. VBN).

3.3 Parameter Accounting

A key property of low-rank factorization is that it does not always reduce the number of parameters. For a matrix of size $d \times d$, the original representation contains d^2 parameters. After factorization, the matrix is stored as two smaller matrices of size $d \times k$ and $k \times d$, for a total of $2dk$ parameters.

This means that compression is achieved only when $2dk < d^2$, or equivalently $k < d/2$. If k exceeds this threshold, the factorized representation will instead become larger than the original matrix. We observe this behavior in our test run of a synthetic 768×768 matrix, where reduction becomes negative once $k > 384$.

This highlights the necessity of selecting an appropriate value of k . We see that smaller values of k reduce memory usage significantly, but also remove more information from the model. Thus, we choose a range of k values to find the ideal value, given the tradeoff between compression and accuracy.

3.4 Targeted Compression of Attention Matrices

We apply truncated SVD to the query, key, and value (Q/K/V) matrices in each attention layer. These matrices are responsible for modeling relationships between words, allowing us to study how compression affects linguistic behavior.

Further, we focus on attention rather than compressing the entire model to isolate its role in encoding syntactic information (Rogers et al., 2020).

SVD is applied after training, and no additional fine-tuning is performed. This allows us to directly evaluate how changing the rank k and removing lower-ranked components affects model

performance.

4 Experimental Setup

Dataset. We use the WSJ_24.POS and WSJ_24.words datasets from the Penn Treebank (Marcus et al., 1993). The data is split into training, development, and test sets, with the test set used only for final evaluation.

Models. We train three transformer models with different parameter sizes: 448K, 1.68M, and 2.25M. The smallest model is trained on 15K sentences, while the larger models are trained on 1K sentences to study the interaction between model size, data availability, and compression.

Compression ranks. We evaluate SVD at ranks $k \in \{1, 2, 4, 8, 16, 24, 32\}$, along with the uncompressed baseline. These values allow us to compare extreme, moderate, and mild compression regimes.

Error analysis. To test our hypotheses, we compute confusion matrices at each compression level and analyze frequent tagging errors. In particular, we examine whether compression increases errors in our three suggested hypotheses

5 Results

5.1 Compression of the 448K Model (15K sentences)

Table 1 reports results for the 448K model trained on 15K sentences. Test accuracy remains within a 0.5-point band across all tested ranks (85.86–86.31%) despite a wide range of parameter disabled rates ranging from 12.5% ($k = 16$) to 94.5% ($k = 1$). Note, the most aggressive compression ($k = 1$, 94.5% of attention parameters disabled) reduces accuracy by only 0.36 points relative to the uncompressed baseline. This result is consistent with Raunak et al. (2019), who observed that simple sentence-level NLU tasks tolerate extremely low-rank compression.

This suggests that the core syntactic knowledge of the model is captured by a small number of dominant components, and that much of the parameter space is redundant.

5.2 Compression of the 1.68M Model (1K sentences)

Table 2 shows that reducing the training corpus to 1K sentences while increasing model size produces a markedly different picture. Baseline accuracy is only 63.82% (down from 86.34% with

SVD k	Param Count	Disabled	Train Loss	Test Loss	Test Acc.	Notes
0	(baseline)	—	0.6780	0.4255	86.34%	uncompressed extreme
1	448	94.53%	0.6804	0.4379	85.98%	
2	896	89.06%	0.6804	0.4379	85.96%	
4	1,792	78.13%	0.7063	0.4416	85.86%	
8	3,584	56.25%	0.9401	0.4317	86.11%	
16	7,168	12.50%	0.7173	0.4353	86.00%	mild

Table 1: 448K-parameter transformer trained on 15K WSJ sentences. Test accuracy is preserved within a 0.5-point band even at 94.5% attention parameter disabling.

SVD k	Param Count	Disabled	Reduction	Train Loss	Test Loss	Test Acc.
0	18,432	0	0%	1.8455	1.2402	63.82%
1	672	17,760	96.35%	1.8465	1.2405	63.81%
2	1,344	17,088	92.71%	1.8465	1.2406	63.81%
4	2,688	15,744	85.42%	1.8465	1.2403	63.79%
8	5,376	13,056	70.83%	2.5255	1.7133	52.67%
16	10,752	7,680	41.67%	2.2162	1.3316	61.77%
24	16,128	2,304	12.50%	2.3179	1.4100	60.30%

Table 2: 1.68M-parameter transformer trained on 1K WSJ sentences. Mid-rank compression ($k = 8$) collapses accuracy to 52.67% while extreme compression ($k = 1$) preserves it.

SVD k	Param Count	Disabled	Reduction	Train Loss	Test Loss	Test Acc.
0	32,768	0	0%	1.7644	1.1849	65.46%
1	896	31,872	97.27%	1.7649	1.1859	65.45%
2	1,792	30,976	94.53%	1.7648	1.1857	65.46%
4	3,584	29,184	89.06%	2.2709	1.4283	57.82%
8	7,168	25,600	78.13%	2.1749	1.2796	62.76%
16	14,336	18,432	56.25%	3.2782	3.0465	11.17%
32	28,672	4,096	12.50%	2.1430	1.2923	62.67%

Table 3: 2.25M-parameter transformer trained on 1K WSJ sentences. Mid-rank compression ($k = 16$) catastrophically collapses test accuracy to 11.17%.

abundant data), but the compressed accuracy is non-monotonic in k . At $k = 1, 2$, and 4 ($\geq 85\%$ disabled) accuracy is essentially contained to a certain range (63.79–63.81%). At $k = 8$ (70.83% disabled), accuracy collapses to 52.67% with a corresponding train-loss spike from 1.85 to 2.53. Higher ranks ($k = 16, 24$) partially recover, yet they also fail to reach the baseline (61.77% and 60.30%).

This non-monotonic pattern suggests that under-trained models rely on a fragile set of features, where removing certain intermediate components is more harmful than removing almost everything.

5.3 Compression of the 2.25M Model (1K sentences)

Table 3 contains the most interesting observation in our study. As seen previously, the lowest ranks ($k = 1, 2$) preserve baseline accuracy (65.45% vs. 65.46%), but at $k = 4$ accuracy already drops to 57.82%, at $k = 8$ it recovers to 62.76%, and at $k = 16$ it catastrophically collapses to 11.17%, ac-

companied by a test-loss explosion from 1.18 to 3.05. Note that this catastrophic collapse is not a monotonic degradation— at $k = 32$, accuracy partially recovers to 62.67%. This pattern suggests an instability at intermediate compression levels. In this range, the model appears to lose important components of its representation without simplifying its behavior, leading to a sharp drop in performance.

6 Error Analysis

Our proposal stated three hypotheses about which linguistic distinctions would degrade most under compression. We now evaluate each in turn against the aggregate confusion counts from the 1K-sentence model at ranks $\{0, 1, 2, 4, 8, 16\}$. Table 4 lists the highest-frequency gold $\rightarrow$ predicted confusion pairs across the union of compression rates (frequency ≥ 10).

Pair (gold $\rightarrow$ pred)	Count	Family
NNP $\rightarrow$ NN	429	Proper / Common
NN $\rightarrow$ NNP	196	Proper / Common
VBN $\rightarrow$ VBD	190	Verb form
NNP $\rightarrow$ JJ	172	Proper / modifier
JJ $\rightarrow$ NN	163	Modifier / noun
JJ $\rightarrow$ NNP	140	Modifier / proper
VB $\rightarrow$ NN	137	Verb / noun
VBP $\rightarrow$ VB	122	Verb form
NN $\rightarrow$ JJ	114	Noun / modifier
VBD $\rightarrow$ VBN	93	Verb form
NN $\rightarrow$ VB	77	Noun / verb
NNPS $\rightarrow$ NNS	77	Plural proper / common
VBG $\rightarrow$ NN	74	Gerund / noun
NNS $\rightarrow$ NN	56	Number
VB $\rightarrow$ VBP	53	Verb form
NNP $\rightarrow$ NNS	51	Proper / plural common
NNS $\rightarrow$ NNP	51	Proper / common

Table 4: Top gold $\rightarrow$ predicted POS confusion pairs (count ≥ 51), aggregated across SVD ranks 0, 1, 2, 4, 8, 16. NNP-related confusions dominate, followed by verb-form (VBN/VBD), modifier (JJ/RB), and noun/modifier confusions. The remaining long tail (17 pairs at counts 40–48) follows the same families.

Prediction 1: Rare and Proper Nouns. This prediction is strongly supported. The most frequent confusion is NNP $\rightarrow$ NN (429), with NN $\rightarrow$ NNP adding another 196 cases (625 total). Additional NNP-related errors appear across multiple tags, leading to an overall total that exceeds 1100. This indicates that compression disproportionately harms rare and specific lexical distinctions, causing the model to default to more frequent noun tags.

Prediction 2: Adjectives vs. Adverbs. We observe consistent confusion between JJ and RB (84 total cases), along with related modifier errors. This supports the hypothesis that compression blurs fine-grained distinctions between modifier categories.

Prediction 3: Context-Dependent Grammar. We find strong evidence of structural degradation through verb-form confusion, with VBN $\rightarrow$ VBD (190) and VBD $\rightarrow$ VBN (93) among the most frequent errors (283 total). These patterns suggest difficulty in resolving context-dependent grammatical relationships. While this highlights a failure in contextual resolution, further token-distance analysis is required to determine if these errors are strictly localized or truly long-distance.

Taken together, these three confusion families—proper/common nouns, adjective/adverb modifiers, and verb-form distinctions—account for the vast majority of compressed-model errors. We can draw the conclusion that while the compressed model

has not lost the general ability to perform POS tagging, its ability for the nuanced and fine-grained distinctions drops sharply.

7 Discussion

Three findings stand out. First, the relationship between compression and accuracy is non-monotonic: extreme compression can be safer than mid-range compression. The collapse at $k = 16$ (11.17%) in the 2.25M model shows that less compression is not always less harmful.

Second, robustness is strongly dependent on training data. While the 448K model trained on 15K sentences tolerates 94.5% parameter removal with minimal loss, the 2.25M model trained on 1K sentences collapsed under the same procedure. So, we can infer that during truncation, under-trained models are more fragile.

Third, compression introduces structured errors. Errors are concentrated in rare nouns, modifier distinctions, and verb-form dependencies. This implies that compression should be evaluated not just by accuracy, but by the types of errors it introduces.

8 Conclusion

We study how truncated SVD compression affects transformer-based POS tagging. Our results show that (i) aggressive low-rank compression can be surprisingly safe, (ii) mid-range compression can cause catastrophic, non-monotonic failure, and (iii) compression primarily harms rare, fine-grained, and context-dependent distinctions.

Future work includes testing larger models (e.g., BERT-base), exploring layer-wise compression, and applying post-compression fine-tuning to mitigate mid-rank collapse.

Limitations

Our experiments are limited to WSJ data and relatively small models (≤ 2.25 M parameters), so results may differ for larger pretrained models. We also did not perform post-compression fine-tuning, which may reduce performance loss. Finally, our analysis relies on aggregate confusion patterns and does not directly measure dependency distance, so long-distance effects are inferred indirectly.

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